A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 03

## Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people<br>**1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **sustainable** → meeting present needs without stopping future generations meeting theirs<br>**Do not credit**<br>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Total impact = population size x the individual (per capita) impact of each person<br>**1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| **01.3** | **Any 1 of 3**<br>1 mark each for any 1 of the 3 points below, in any order, maximum 1.<br>**1** Slash and burn agriculture<br>**1** Fishing<br>**1** Discharge of sewage / biodegradable pollutants<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **sustainable** → meeting present needs without stopping future generations meeting theirs<br>• **biodegradable** → broken down by decomposers<br>**Do not credit**<br>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1. | |
| **02.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1** 2 400 / 2 000<br>**1** = 1.2<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• **food conversion ratio** → the feed needed per unit of food produced<br>**Do not credit**<br>• A bare wrong answer with no working. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Carnivorous species have a high market value, so they are the most profitable to farm, and consumers in wealthy countries prefer them<br>**1** They are fed on fishmeal and fish oil made from wild-caught fish, so farming them still depends on catching wild fish and does not take the pressure off marine stocks<br>**1** Several tonnes of wild fish may be needed per tonne of salmon, so the extra trophic level means the total amount of fish protein available falls rather than rises<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **aquaculture** → fish or shellfish farming<br>• **trophic level** → the feeding level in a food chain<br>**Do not credit**<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts | 3 |
| **02.3** | **Any 4 of 6**<br>1 mark each for any 4 of the 6 points below, in any order, maximum 4.<br>**1** Uneaten feed and faeces fall through the cages and accumulate on the sea bed, where their decomposition removes oxygen and smothers the benthic community<br>**1** The nutrients released cause local eutrophication and can trigger algal blooms<br>**1** Parasites, especially sea lice, build up in the dense stock and spread to wild salmon and sea trout passing the cages<br>**1** Antibiotics, pesticides used to treat sea lice and antifouling chemicals from the nets disperse into the water and affect non-target species<br>**1** Escaped farmed fish interbreed with wild populations, diluting the locally adapted gene pool, and compete with them for food and spawning sites<br>**1** Predators such as seals and cormorants attracted to the cages are shot or trapped<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **eutrophication** → nutrient enrichment of water leading to algal growth<br>• **gene pool** → the range of alleles / genes present in a population<br>**Do not credit**<br>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.<br>• 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.4** | **Any 3 of 5**<br>1 mark each for any 3 of the 5 points below, in any order, maximum 3.<br>**1 Integrated multi-trophic aquaculture:** seaweeds and filter feeders such as mussels are grown alongside the cages and take up the dissolved nutrients and particulate waste as their own food<br>**1** Siting cages where the current is strong, and leaving sites fallow between cycles, so wastes are dispersed and the sea bed recovers<br>**1** Lowering stocking density and vaccinating the fish, so less medication is needed, and using cleaner fish such as wrasse to remove sea lice instead of pesticides<br>**1** Stronger, better maintained nets and predator exclusion nets, so fewer fish escape and predators do not have to be killed<br>**1** Replacing part of the fishmeal in the feed with plant or insect protein, reducing the demand on wild fish stocks<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **aquaculture** → fish or shellfish farming<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts | 3 |
| **02.5** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Farming carnivorous species still depends on wild fish for feed, so expanding it increases rather than reduces the catch of wild fish<br>**1** Suitable sheltered coastal sites are limited, and they compete with other uses and with sensitive habitats such as mangroves, which shrimp farming has already destroyed on a large scale<br>**1** Some valuable species, such as tuna, cannot yet be bred reliably in captivity, so farms are stocked with wild-caught young fish and the wild population is still exploited<br>**1** The local pollution, disease and escape problems mean that simply scaling the industry up multiplies its impacts<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **aquaculture** → fish or shellfish farming | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-359; AQA spec 3.5.2.3. | |
| **03.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1** B - radio waves do not travel through seawater<br>– acoustic or satellite tags transmit position | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.2** | **2 marks per point**<br>2 marks for each of the 2 points below; award 1 mark where a point is only partly made.<br>**2** Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved<br>**2** Radio / GPS tags or collars<br>– leg rings / wing tags on birds<br>– camera traps<br>– DNA from droppings to confirm identity and diet<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **captive breeding** → breeding in captivity to increase numbers for release<br>**Do not credit**<br>• captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6. | |
| **04.1** | **2 marks per point**<br>2 marks for each of the 3 points below; award 1 mark where a point is only partly made.<br>**2** Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands<br>– HEP schemes alter river levels flooding sandbank nests of turtles / birds<br>– species with narrow tolerance (specialists) affected first<br>– sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade<br>**2** Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae<br>**2** Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **abiotic** → non-living / physical factors<br>• **abstraction** → water taken out of a river, lake or aquifer<br>• **HEP** → electricity generated from falling water<br>**Do not credit**<br>• carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen<br>• a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation | 6 |
| **04.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Reduces light penetration so submerged plants cannot photosynthesise<br>**1** Blocks the gills of filter feeders such as bivalve molluscs<br>– sediment smothers the bed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Rare species often have very specialised habitat requirements / narrow ranges of tolerance<br>**1** A small change moves conditions outside their range while generalists survive<br>– small populations cannot recover<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **abiotic** → non-living / physical factors<br>• **tolerance** → the range of conditions in which a species can survive | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2. | |
| **05.1** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** The size of the population, and the number of mature individuals able to breed<br>**1** The rate of change of the population: how fast numbers have fallen, usually over the last ten years or three generations<br>**1** The geographical range the species occupies and how far it is fragmented or dispersed, since a small or divided range makes extinction more likely<br>**1** The quality and area of the remaining habitat, and whether the threats acting on the species are continuing<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. | 3 |
| **05.2** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1** (12 000 - 2 100) / 12 000 x 100<br>**1** = 82.5% (accept 82 - 83%)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (%) is already printed on the answer line.<br>**Do not credit**<br>• Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.<br>• A bare wrong answer with no working. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.3** | **Any 3 of 5**<br>1 mark each for any 3 of the 5 points below, in any order, maximum 3.<br>**1** It is a keystone species, so losing it would change the whole community and many other species would be lost with it<br>**1** It is endemic to the area, so if it is lost there it is lost everywhere – a species found in many countries can be conserved elsewhere<br>**1** It is a flagship or charismatic species that attracts public interest and funding, which also protects everything else living in its habitat<br>**1** It is a genetic resource with known or likely value: a wild relative of a crop, or a source of a medicine<br>**1** Conservation is practical and likely to succeed: the threats can be removed, and suitable habitat remains for a reintroduction<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **keystone species** → a species whose loss changes the whole community<br>• **endemic** → found naturally only in that one place<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. | 3 |
| **05.4** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Appendix I lists species threatened with extinction, for which commercial international trade is banned and only exceptional permits such as scientific exchange are issued<br>**1** Appendix II lists species that are not yet threatened with extinction but may become so unless trade is controlled, so trade is allowed under licence and monitored<br>**1** Where the threat is demand in distant wealthy markets, cutting off the market removes the incentive to poach, and the agreement acts in every signatory country rather than only where the habitat is protected<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **CITES** → the treaty that controls international trade in endangered species<br>**Do not credit**<br>• 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.5** | **Any 4 of 6**<br>1 mark each for any 4 of the 6 points below, in any order, maximum 4.<br>**1** Sites of Special Scientific Interest are notified for their wildlife or geology, and damaging operations on them need consent from the conservation agency<br>**1** National Nature Reserves and Local Nature Reserves are owned or managed for conservation, with management plans and wardens<br>**1** The Wildlife and Countryside Act makes it an offence to kill, injure or take listed species, or to damage their breeding sites and resting places<br>**1** Special Areas of Conservation and Special Protection Areas, and Ramsar sites for wetlands, give international protection to habitats and to migratory birds<br>**1** Green belts, National Parks and Areas of Outstanding Natural Beauty restrict development and so protect habitat indirectly<br>**1** Agri-environmental schemes pay farmers to manage hedgerows, field margins and meadows for wildlife on land that is still farmed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-32, 38-39; AQA spec 3.1.2.3.1-3.1.2.3.2. | |
| **06.1** | **Any 2 of 6**<br>1 mark each for any 2 of the 6 points below, in any order, maximum 2.<br>**1** Hydrological cycle<br>**1** Carbon cycle<br>**1** Nitrogen cycle<br>**1** Atmospheric circulation<br>**1** Thermohaline circulation<br>**1** Photosynthesis<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **Thermohaline** → the ocean conveyor driven by temperature and salinity<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration | 2 |
| **06.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Enzymes reduce the activation energy of reactions<br>**1** So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.3** | **Any 1 of 3**<br>1 mark each for any 1 of the 3 points below, in any order, maximum 1.<br>**1** Haber process (nitrogen fixation)<br>**1** Incineration of wastes (decomposition)<br>**1** Many manufacturing processes<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds<br>• **Incineration** → burning waste, usually with energy recovery | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2. | |
| **07.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Upwelling currents bring nutrients from the seabed to surface waters<br>**1** 24-hour daylight in summer allows a huge phytoplankton bloom<br>**1** Algae also grow under sea ice where krill shelter and feed – krill support most of the food web<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **07.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Krill is the main food of baleen whales, penguins, seals and fish so over-harvesting removes food from most of the web<br>**1** Toothfish overfishing removes the food of sperm whales and elephant seals<br>**1** Long-lining for toothfish catches and drowns albatrosses on the hooks (by-catch)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 3 |
| **07.3** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Ice on land stores 70% of the Earth's fresh water, keeping sea levels low<br>**1** High albedo of ice reflects sunlight, limiting warming (reduction in ice area causes further warming)<br>**1 Carbon sequestration:** carbon in algae eaten by krill sinks to the seabed in krill faeces<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back<br>**Do not credit**<br>• 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.4** | 1 mark for the point below.<br>**1** Stronger winds spreading ice<br>– more snowfall from warmer, moister air freezing onto slush<br>– fresh meltwater from land forming a surface layer that freezes more easily (long-term trend still expected to be a decline)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 67-71; AQA spec 3.1.2.3.4. | |
| **08.1** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis<br>**1** In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May)<br>**1** The bloom uses up nutrients (falling to 10 by July)<br>– dead plankton sink out of the photic zone carrying nutrients to the seabed<br>**1** So nutrients become limiting in summer despite high light<br>– a smaller autumn bloom follows when storms return nutrients while light is still adequate<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 4 |
| **08.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced<br>**1** Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **limiting factor** → the factor in shortest supply that holds the rate or population back | 2 |
| **08.3** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone<br>**1** Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments<br>**1** Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **08.4** | 1 mark for the point below.<br>**1** Phosphates have low solubility and no atmospheric reservoir<br>– they are only added by river runoff or upwelling and are lost to sediments<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1. | |
| **09.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** All greenhouse gases expressed as CO2 equivalent (allowing for different warming strengths)<br>**1** Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis)<br>**1** A fair share of the infrastructure: roads, car parks, signals, street lighting<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **09.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels<br>**1** Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **biodiversity** → the variety of life: habitat, species and genetic diversity | 2 |
| **09.3** | **1 mark each**<br>1 mark for each of the 5 points below.<br>**1** Hardy, disease-resistant Boran cattle<br>– regular tick dipping reduces ticks for wild herbivores too<br>**1** Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth<br>**1** Electric fences keep game in and poachers out<br>– biological corridors let animals (except rhinos) move to other reserves<br>– problem elephants relocated<br>**1 Monitoring:** radio-collared lions, drones, camera traps, track surveys<br>– endangered species (Grevy's zebra, northern white rhino) protected<br>**1 Community links:** ecotourism income funds health, education and water projects<br>– local employment<br>– cattle breeding help for local farmers<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 5 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.1** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **In-situ:** species remain in natural conditions with the whole community and inter-species relationships<br>– protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers)<br>• **Limits of in-situ:** external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations<br>• **Ex-situ:** captive breeding when in-situ is impossible or insufficient<br>– selection criteria<br>– difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation)<br>– techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning)<br>– seed banks<br>– successes (Hawaiian goose, Iberian lynx, red kite, large blue)<br>• **Limits of ex-situ:** expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring)<br>• **Judgement:** ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to<br>– both needed, in-situ fundamental<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content.<br>• **abiotic** → non-living / physical factors<br>• **biotic** → living factors / effects of other organisms | 9 |
| | • **plagioclimax** → a community held back from its climax by human activity<br>**Do not credit**<br>• captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well | |
| | **Model answer - Level 3 (8 marks)** | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | In-situ conservation keeps a species in its natural habitat with the whole community it depends on - food species, pollinators, predators and the abiotic conditions it is adapted to. It is delivered through protected areas such as SSSIs, NNRs, SACs, SPAs, Ramsar sites and Marine Conservation Zones, land ownership by conservation bodies, agri-environment schemes and habitat creation and management. Its weaknesses are that external threats such as pollution and climate change cross reserve boundaries, enforcement and cost, conflicts with other land uses, and the fact that many populations are already small and fragmented. | |
| | Ex-situ captive breeding is used when in-situ protection is impossible or insufficient. It is hard: animals need enough space, the right food, social groupings and breeding triggers, and small founder populations risk inbreeding and hybridisation, which studbooks, artificial insemination and embryo transfer help to manage. It is expensive, works for few species, and captive animals may lose wild behaviour; release succeeds only where suitable habitat exists and with soft release and post-release monitoring. | |
| | The two are not alternatives. Ex-situ methods are a temporary safeguard that can rescue a species from extinction, as with the Arabian oryx, but they only succeed if there is protected habitat to return to. In-situ conservation is therefore fundamental and ex-situ a support for it; the relative importance depends on how close to extinction the species is. | |
| | **Weaker answer - Level 2 (5 marks)** | |
| | In-situ conservation means protecting species in their natural habitat, for example in nature reserves and national parks. This is good because the species stays in the wild with the other species it needs. Ex-situ conservation means captive breeding in zoos and then releasing the animals back into the wild. This can save species that are nearly extinct but it is expensive and animals may not survive when released. Both methods are important and are often used together. | |
| | **What makes the difference** | |
| | • Level 2 defines both approaches correctly but Level 3 gives specific mechanisms (types of protected area, breeding difficulties, release methods) and named examples. | |
| | • Level 3 weighs the limitations of each approach explicitly; Level 2 gives one limitation for ex-situ only. | |
| | • 'Both are important' is not a judgement about *relative* importance; Level 3 explains why in-situ is fundamental and when ex-situ matters most. | |
| **10.2** | 1 mark for the point below.<br>**1** Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.1 | **Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Setting priorities:** IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal<br>• **Legislation and protocols:** Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement<br>• **Captive breeding and release:** selection, difficulties, techniques, release site criteria, hard / soft release, monitoring<br>– seed banks<br>• **Habitat conservation:** creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions<br>• Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts<br>• Monitoring and new technologies informing decisions<br>• **Evaluation:** examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers)<br>– cost, politics, climate change<br>– judgement on the most effective combination | 25 |
| | *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3. | |
| | **Model answer - Level 5 (23 marks)** | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | Biodiversity is lost through direct exploitation, habitat destruction, introduced species, pollution and changes to abiotic factors, so its conservation needs methods that address each cause, at scales from a single species to international law. Their effectiveness varies with the threat, the species and the resources available. | |
| | Deciding what to conserve comes first. The IUCN Red List uses population size, area, decline rate and probability of extinction to prioritise species, and endemic, keystone, flagship and evolutionarily distinct species receive weight because their loss has wider consequences. This is effective at directing scarce resources, but it favours well-studied vertebrates over invertebrates and plants. | |
| | Legislation and agreements protect species and habitats. In the UK the Wildlife and Countryside Act, SSSIs, NNRs and Marine Conservation Zones, and in Europe SACs, SPAs and the Habitats Directive, restrict damaging activities; internationally CITES controls trade in threatened species, the IWC regulates whaling, the Ramsar Convention protects wetlands and the Rio Convention on Biological Diversity commits nations to conservation. Their effectiveness depends on enforcement: CITES has reduced trade in ivory and rhino horn but poaching continues where enforcement is weak; the IWC moratorium allowed some whale populations to recover but Japan and Norway continued whaling; designations protect against development but not against pollution or climate change crossing boundaries. | |
| | In-situ habitat conservation and management are the most fundamental methods because they keep species with the community and abiotic conditions they depend on. Management uses ecological understanding: maintaining plagioclimaxes such as chalk grassland and heathland by grazing or burning, coppicing and pollarding woodland, controlling water levels, light and grazing, designing reserves with the right area, shape, biological corridors and buffer zones, and creating new habitats such as wetlands in worked-out gravel pits. Case studies show it works - coppiced woodland restored dormice and butterflies, wetland creation increased bitterns - but reserves are often small and fragmented and need permanent funding. | |
| | Ex-situ methods rescue species for which in-situ protection has failed. Captive breeding faces problems of space, diet, breeding triggers, social groups and small gene pools, addressed with studbooks, artificial insemination, embryo transfer and cryopreservation; seed banks such as the Millennium Seed Bank store plant genetic diversity cheaply. The Arabian oryx, California condor and pool frog show captive breeding can bring species back, but only where protected habitat exists for hard or soft release with post-release monitoring, and it is expensive and works for few species. | |
| | Ecological monitoring - image databases, eDNA, tracking, satellites - measures whether any of this is working and has improved the targeting of effort. Overall, legislation is effective when enforced, habitat management is the most effective and widely applicable method, and captive breeding is a last resort that succeeds only in combination with habitat protection. The Living Planet Index continues to fall, however, which shows that conservation methods are winning individual battles while the drivers of loss - population, consumption, land-use change and climate change - remain unaddressed. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **What makes the difference** | |
| | • The essay is structured by **type of method** and evaluates the effectiveness of each in turn, with named examples for both successes and failures. | |
| | • It draws on several specification areas (biodiversity, conservation legislation, habitat management, captive breeding, ecological processes, monitoring) - the synoptic breadth Level 5 requires. | |
| | • The conclusion ranks the methods and then steps back to the bigger picture (Living Planet Index, underlying drivers) - a supported judgement rather than a summary. | |
| | • A Level 3 essay would describe the same methods accurately but with fewer examples, little evaluation of enforcement or cost, and a conclusion that all methods are important. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.2 | **Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Physical features:** mass / gravity retaining an atmosphere and giving pressure for liquid water<br>– distance from the Sun and the Goldilocks zone<br>– speed of rotation reducing temperature extremes<br>– tilt producing seasons<br>– magnetic field deflecting the solar wind<br>– the Moon stabilising the orbit and producing tides<br>• **Importance of liquid water:** solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer<br>• **Development of life:** Archaea at vents<br>– photosynthesis<br>– oxygen reacting with iron then accumulating<br>– ozone layer allowing colonisation of land<br>– aerobic organisms<br>• **Life changing conditions:** carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases | 25 |
| | • **Methods of studying past conditions:** proxy data, isotope ratios, ice cores, Mg:Ca ratios<br>– understanding of continental drift and ocean currents<br>• Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1. | |

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